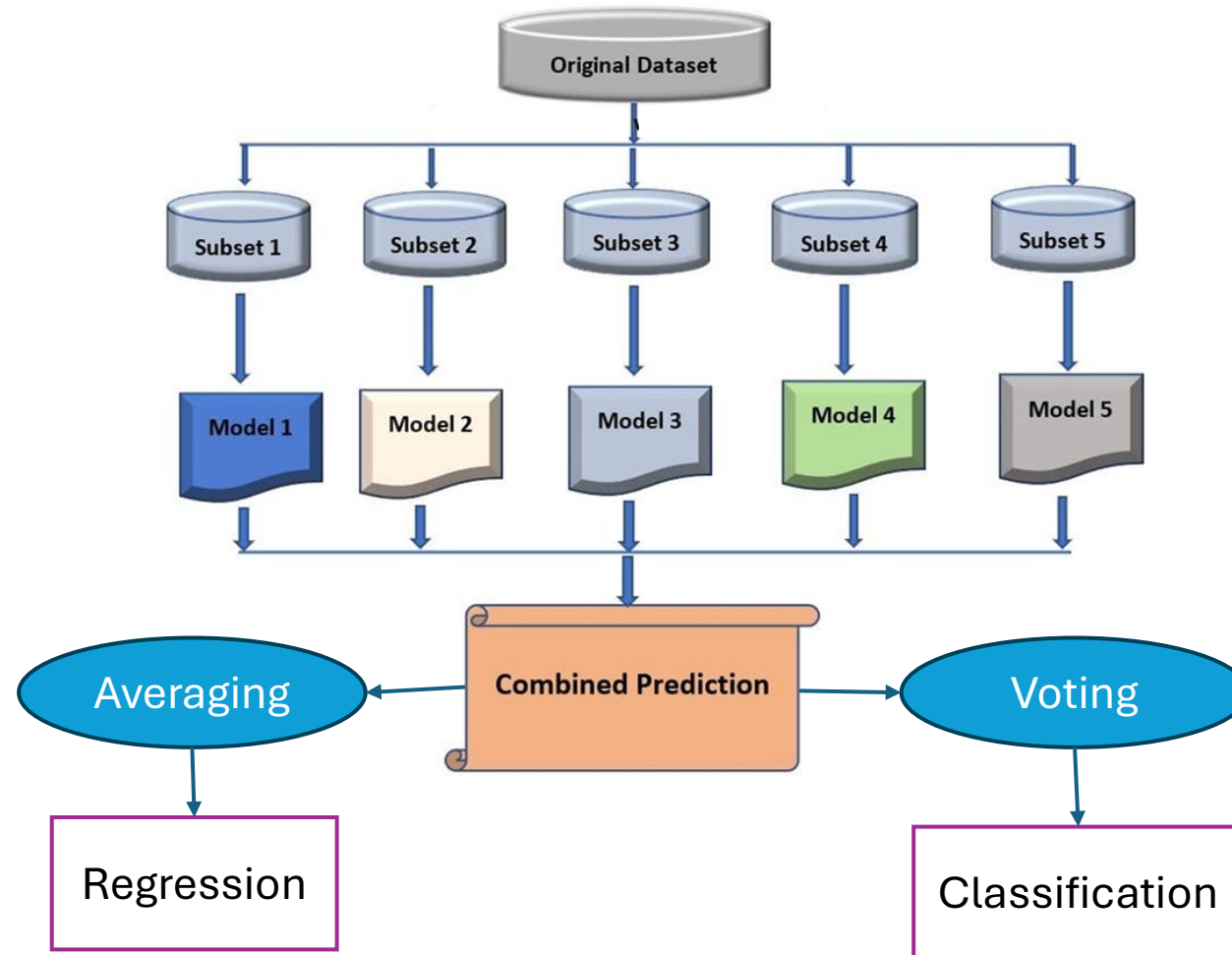


Ensemble Techniques

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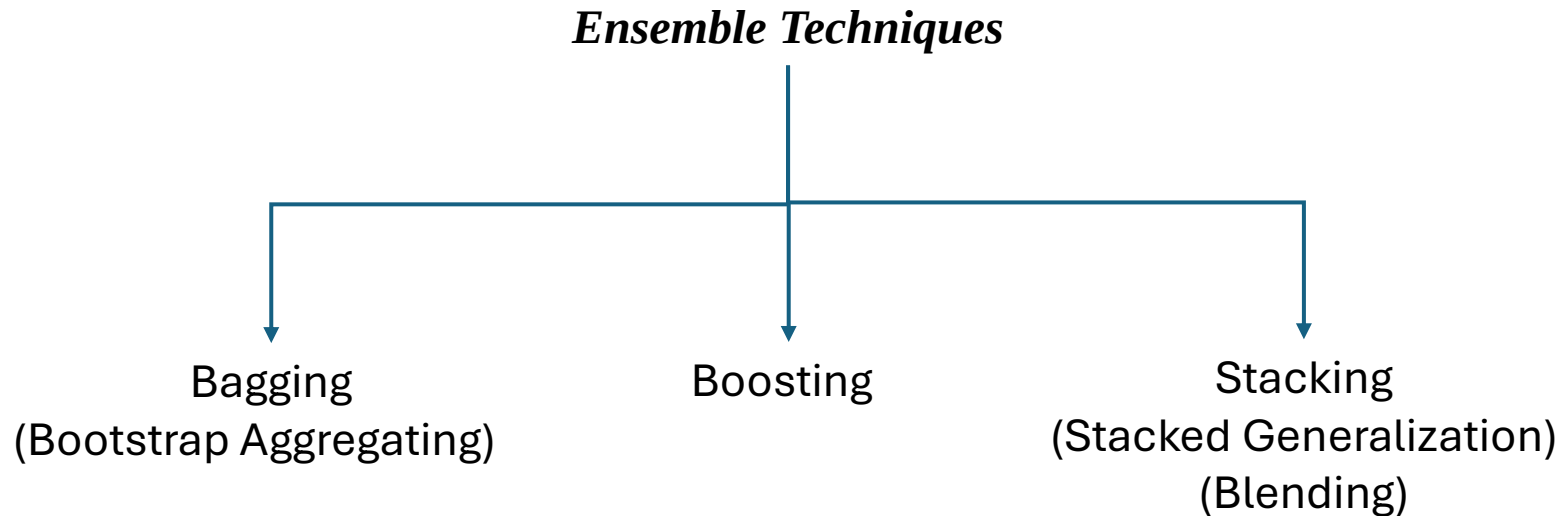
Ensemble Techniques

Ensemble learning is a machine learning technique that combines multiple models to improve performance and robustness. Instead of relying on a single model, ensemble methods use multiple models (weak learners) and aggregate their predictions to obtain better accuracy and generalization.



Ensemble Techniques

Ensemble learning is a machine learning technique that combines multiple models to improve performance and robustness. Instead of relying on a single model, ensemble methods use multiple models (weak learners) and aggregate their predictions to obtain better accuracy and generalization.



Ensemble Techniques

Bagging

Concept: Bagging creates multiple independent models by training them on different subsets of the dataset. The final prediction is obtained by averaging (for regression) or majority voting (for classification).

Purpose: Reduces variance by training multiple models on different random subsets of the data.

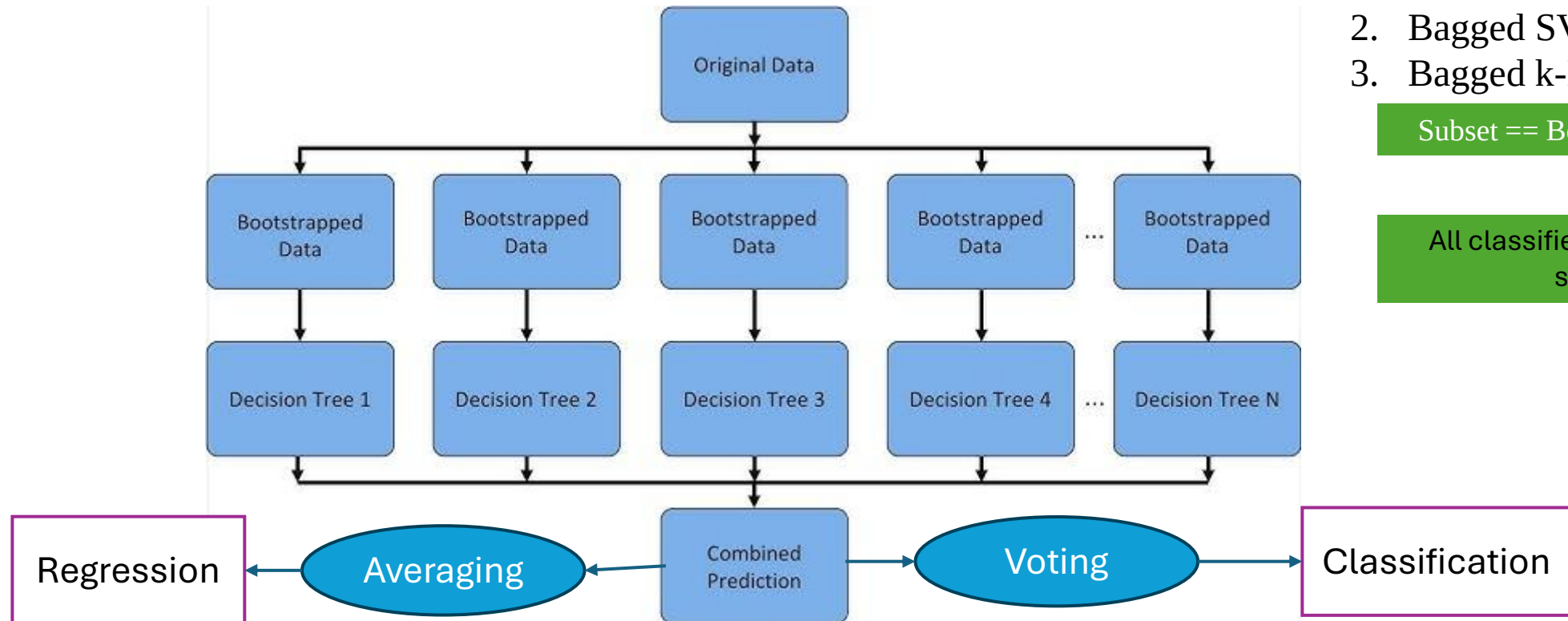
Example: Random Forest, where multiple decision trees are trained on different random subsets of the data and their predictions are averaged (for regression) or voted on (for classification).

Some common bagging models

1. Extra Trees
2. Bagged SVM
3. Bagged k-NN

Subset == Bootstrapped Data

All classifiers here are the same.



Ensemble Techniques

Boosting

Concept: Reduces bias by sequentially training models, where each new model focuses on correcting errors made by previous models. The final prediction is a weighted sum of all models.

Purpose: Reduces bias and improves predictive accuracy.

1. **Example Algorithms: AdaBoost (Adaptive Boosting)** – Assigns higher weights to misclassified samples.
2. **Gradient Boosting** – Uses gradient descent to optimize the loss function.
3. **XGBoost, LightGBM, CatBoost** – More optimized versions of boosting.

Ensemble Techniques

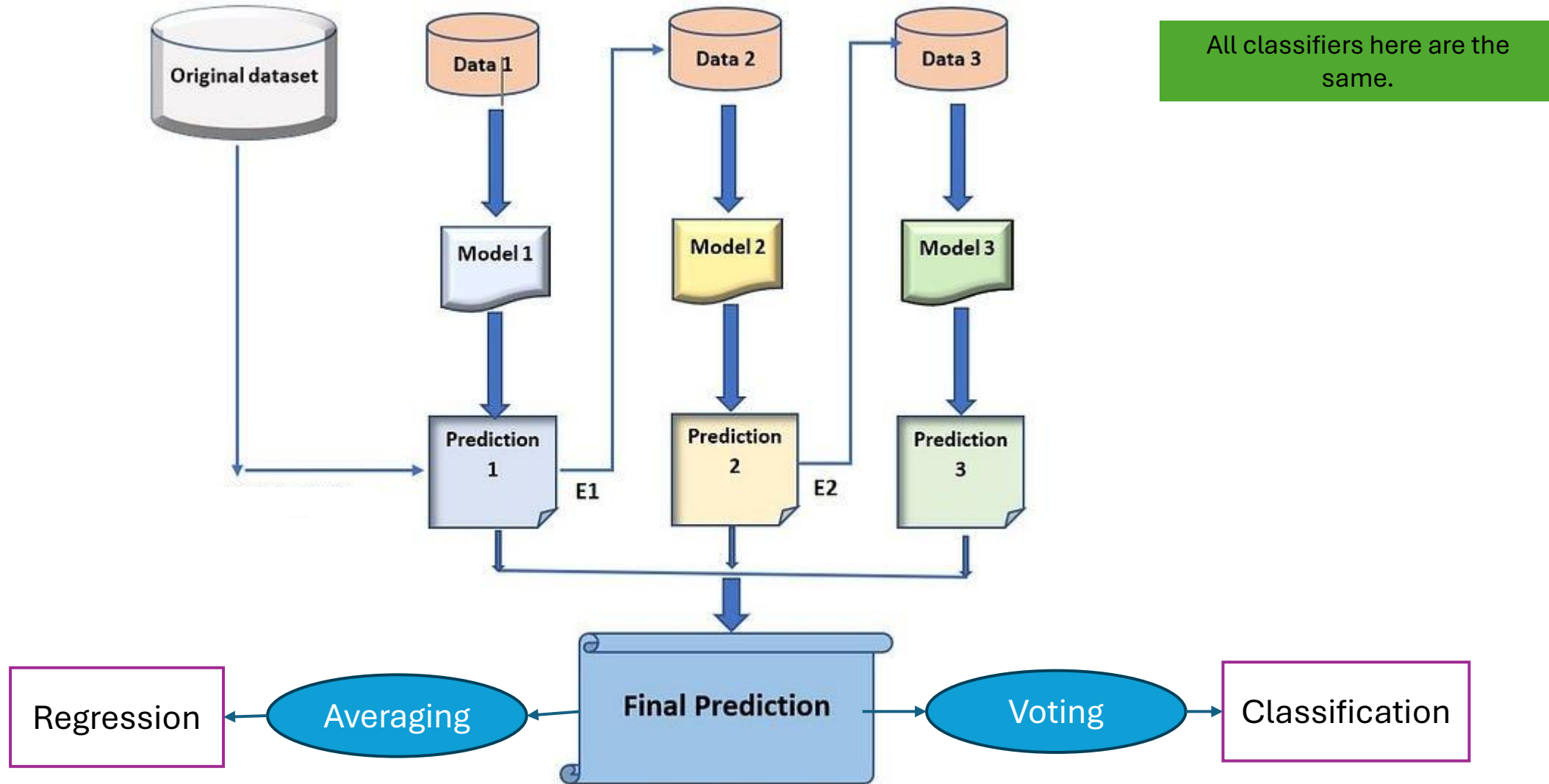
Boosting

Steps in Boosting:

1. A subset is created from the original dataset.
2. Initially, all the data points are given equal weights, i.e. all the data points are considered equally likely for random sampling.
3. A base model is trained on the randomly sampled subset.
4. Errors are calculated using actual values and predicted values.
5. The observations which are incorrectly predicted, are given higher weights. These observations will be picked first on the next data sampling for the next model training because of their high weight, and then the rest are picked randomly if needed.
6. Another model will be created and predictions are made on the dataset. (This model tries to correct the errors from the previous model).
7. Similarly, multiple models will be created with updated weights, each correcting the errors of the previous model.
8. When all training processes of weak learners are done, they are combined together to form a strong learner. The final model (strong learner) is the weighted mean of all the models (weak learners).

Ensemble Techniques

Boosting



Ensemble Techniques

Stacking (Stacked Generalization)

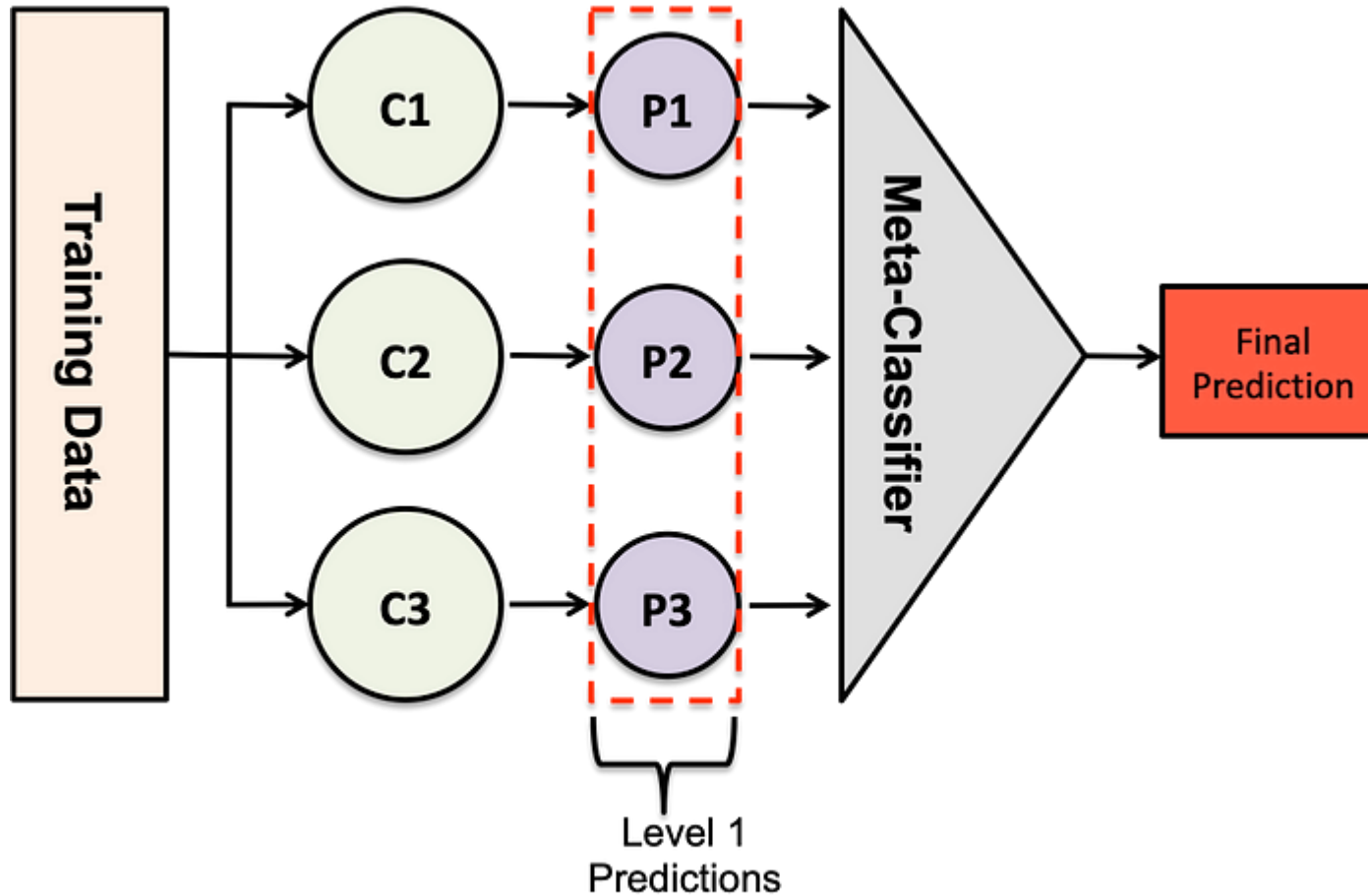
Concept: Stacking combines multiple models (diverse base learners) and trains a **meta-model** (or blender) on their outputs to make the final prediction. The base models learn from the dataset, and their outputs become the input features for the meta-model.

Purpose: Leverages the strengths of multiple models to improve overall performance.

Example: Using Logistic Regression as a meta-model to combine predictions from Random Forest, XGBoost, and SVM.

Ensemble Techniques

Stacking (Stacked Generalization)



* C1, C2, and C3 are considered level 1 classifiers.

All classifiers
here are
different.

Ensemble Techniques

Stacking (Stacked Generalization)

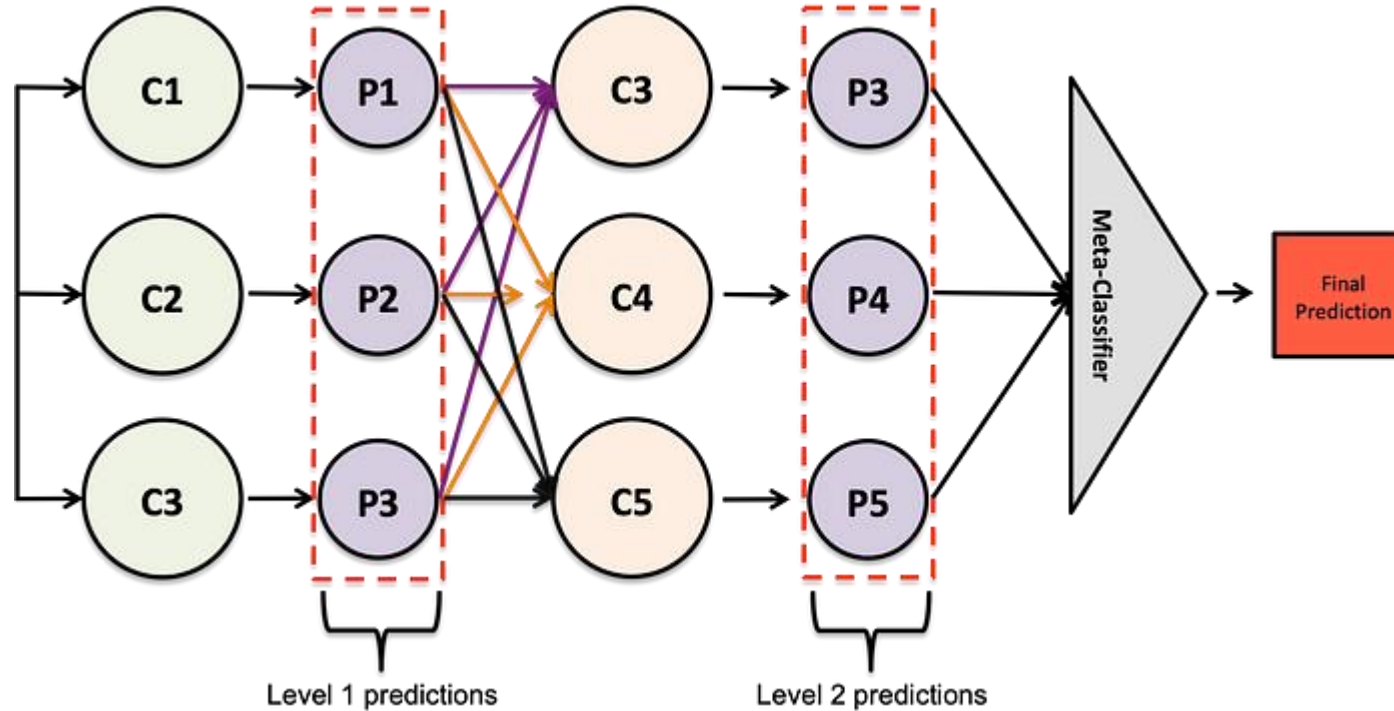


Fig: Stacking classifier with two layers of classifiers

All classifiers
here are
different.

Ensemble Techniques

Comparative Analysis

Technique	Approach	Reduces	Example Algorithm
Bagging	Parallel	Variance	Random Forest
Boosting	Sequential	Bias	AdaBoost, XGBoost
Stacking	Parallel + Meta-learner	Bias & Variance	Stacked Ensemble

Ensemble Techniques

Bias and Variance

Predicting House Prices

We have a dataset of 5 houses with their square footage and prices.

Given Dataset

Square Feet (X)	Actual Price (Y)
1000	150
1500	200
2000	250
2500	310
3000	400

We will fit three different models:

1. High Bias Model (Underfitting) → Simple Linear Regression
2. High Variance Model (Overfitting) → High-degree Polynomial Regression
3. Optimal Model (Good Balance) → Moderate-degree Polynomial Regression

Ensemble Techniques

Bias and Variance

Simple Linear Regression → High Bias Model (Underfitting Model)

We fit a simple regression model: $Y=0.1X+50$

Square Feet (X)	Actual Price (Y)	Predicted Price $Y=0.1X+50$	Error $(Y - \text{Predicted } Y)^2$
1000	150	150	$(150 - 150)^2 = 0$
1500	200	200	$(200 - 200)^2 = 0$
2000	250	250	$(250 - 250)^2 = 0$
2500	310	300	$(310 - 300)^2 = 100$
3000	400	350	$(400 - 350)^2 = 2500$

Total Mean Squared Error (MSE)

$$\text{MSE} = \frac{0+0+0+100+2500}{5} = 520$$

High Bias: The model is **too simple**, missing the real pattern → **Underfitting**.

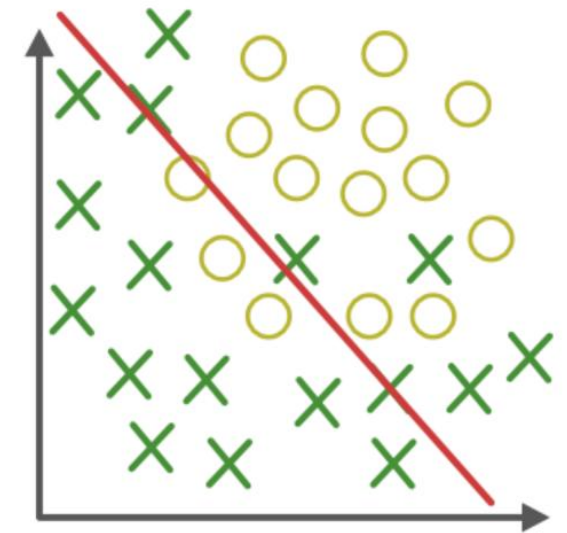


Fig: Underfitting

Ensemble Techniques

Bias and Variance

Higher degree Polynomial Regression → High Variance Model (Overfitting Model)

We fit a high degree polynomial regression model:

$$Y = (6.67 \times 10^{-12}) X^4 - (3.33 \times 10^{-8}) X^3 + (5.83 \times 10^{-5}) X^2 + (5.83 \times 10^{-2}) X + 60$$

Square Feet (X)	Actual Price (Y)	Predicted Price Y	Error (Y - Predicted Y) ²
1000	150	149.97	0.0009
1500	200	200.00438	1.9141
2000	250	250.12	0.0144
2500	310	310.35938	0.1292
3000	400	400.77	0.5929

Total Mean Squared Error (MSE)

$$\text{MSE} = 0.147473906$$

High Variance: The model **perfectly fits** the training data but might fail on new data → **Overfitting**.

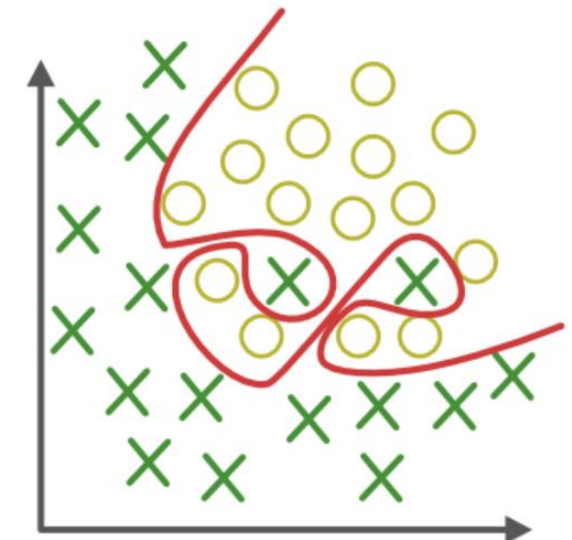
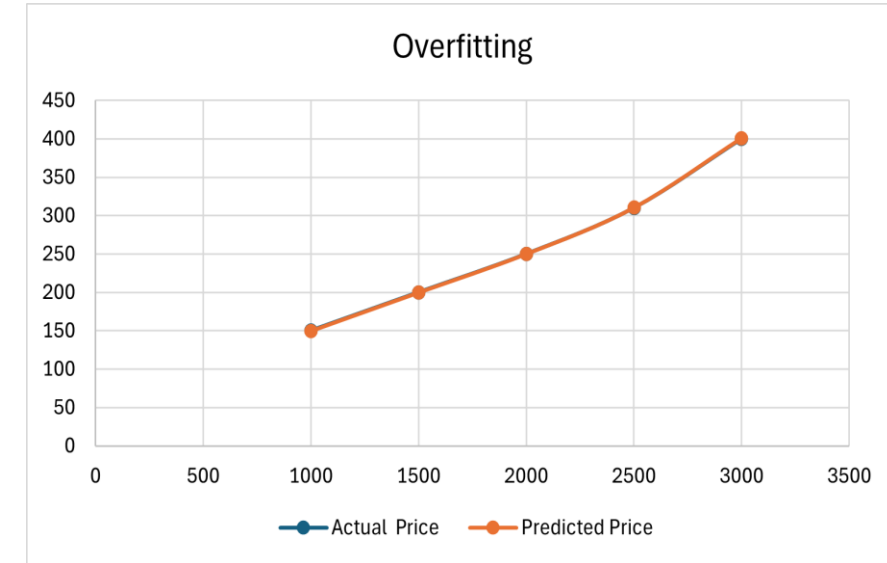


Fig: Overfitting

Ensemble Techniques

Bias and Variance

Moderate degree Polynomial Regression → Good Bias-Variance Tradeoff Model (Optimal Model)

We fit a high degree polynomial regression model:

$$Y = (2.57 \times 10^{-5}) X^2 + 0.019X + 105$$

Square Feet (X)	Actual Price (Y)	Predicted Price Y	Error (Y - Predicted Y) ²
1000	150	149.7	0.09
1500	200	191.325	75.255625
2000	250	245.8	17.64
2500	310	313.125	9.765625
3000	400	393.3	44.89

Total Mean Squared Error (MSE)

$$\text{MSE} = 29.52825$$

Optimal Balance: The model captures patterns without overfitting → **Best Tradeoff.**

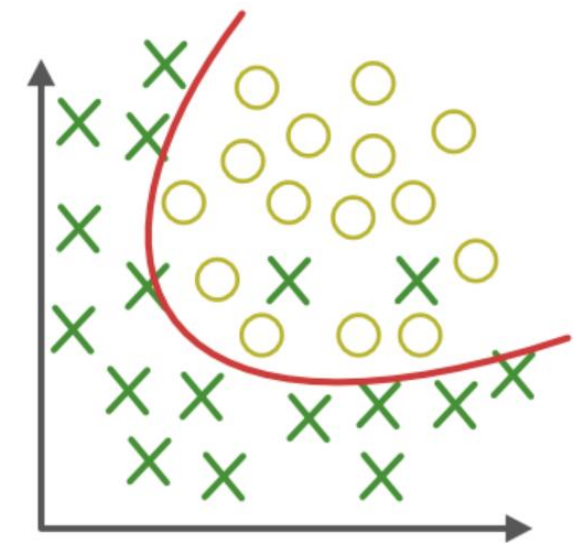


Fig: Overfitting

Ensemble Techniques (AdaBoost/Adaptive Boosting)

Simple Dataset

Sample	X1	X2	Y(Target)
1	2	3	+1
2	3	1	+1
3	1	2	-1
4	4	5	+1
5	2	1	-1
6	5	4	-1

The target (Y) has **binary classes** $\{+1, -1\}$, so we can apply AdaBoost.

Step 1: Initialize Sample Weights

Initial Weight (w_i) = $\frac{1}{N}$ ----- (1)

N represents the total number of samples in the dataset. For this dataset, N=6.

Sample	X1	X2	Y(Target)	Initial Weight (w_i)
1	2	3	1	0.1667
2	3	1	1	0.1667
3	1	2	-1	0.1667
4	4	5	1	0.1667
5	2	1	-1	0.1667
6	5	4	-1	0.1667

Step 2: Choose a base classifier and number of repetition

In this case, a Decision Tree has been selected as the base classifier, and the number of repetitions has been set to 5.

Step 3: Train First Weak Classifier (**Iteration 1**)

Let's use a simple **Decision Stump** (one-feature decision tree). We'll choose a decision rule like:

➤ **Rule:** If $X_1 \geq 3$, predict 1; else predict -1.

Sample	X1	Y(Target)	Prediction	Correct?
1	2	1	-1	×
2	3	1	1	✓
3	1	-1	-1	✓
4	4	1	1	✓
5	2	-1	-1	✓
6	5	-1	1	×

Ensemble Techniques (AdaBoost/Adaptive Boosting)

Misclassified Samples: 1 and 6

Total Error (ϵ) Calculation:

$$\epsilon = \sum \text{Weight of misclassified samples}$$

$$\epsilon = w_1 + w_6 = 0.1667 + 0.1667 = 0.3333$$

Compute Classifier Weight (α)

$$\alpha = \frac{1}{2} \ln \left(\frac{1 - \epsilon}{\epsilon} \right)$$

$$\alpha = \frac{1}{2} \ln \left(\frac{1 - 0.3333}{0.3333} \right)$$

$$\alpha = \frac{1}{2} \ln \left(\frac{0.6667}{0.3333} \right) = \frac{1}{2} \ln(2) = \frac{1}{2}(0.693) = 0.3466$$

Update Sample Weights

New weight formula:

$$w_{\text{new}} = w \times e^{\pm\alpha}$$

- Correctly classified samples ($w_{\text{new}} = w \times e^{-\alpha}$)
- Misclassified samples ($w_{\text{new}} = w \times e^{\alpha}$)

1. Compute weight update factors:

- $e^{-\alpha} = e^{-0.3466} = 0.7071$
- $e^{\alpha} = e^{0.3466} = 1.4142$

Ensemble Techniques (AdaBoost/Adaptive Boosting)

2. Compute new weights:

Sample	Old Weight	New Weight Formula	Updated Weight
1 (✗)	0.1667	0.1667×1.4142	0.2356
2 (✓)	0.1667	0.1667×0.7071	0.1178
3 (✓)	0.1667	0.1667×0.7071	0.1178
4 (✓)	0.1667	0.1667×0.7071	0.1178
5 (✓)	0.1667	0.1667×0.7071	0.1178
6 (✗)	0.1667	0.1667×1.4142	0.2356

3. Normalize new weights so they sum to 1:

$$Z = 0.2356 + 0.1178 + 0.1178 + 0.1178 + 0.1178 + 0.2356 = 1.000$$

Thus, final updated weights remain the same.

Ensemble Techniques (AdaBoost/Adaptive Boosting)

Iteration 2: Train Second Weak Classifier

Now, using the **updated weights**, we train another weak classifier, focusing more on samples **1 and 6**, which had **higher weights**.

- **New Rule:** If $X_2 \geq 2$, predict $+1$; else predict -1 .

Sample	X_2	True Y	Prediction	Correct?
1	3	+1	+1	✓ Correct
2	1	+1	-1	✗ Wrong
3	2	-1	+1	✗ Wrong
4	5	+1	+1	✓ Correct
5	1	-1	-1	✓ Correct
6	4	-1	+1	✗ Wrong

Ensemble Techniques (AdaBoost/Adaptive Boosting)

- Misclassified Samples: 2, 3, 6

$$\epsilon = w_2 + w_3 + w_6 = 0.1178 + 0.1178 + 0.2356 = 0.4712$$

$$\alpha = \frac{1}{2} \ln \left(\frac{1 - 0.4712}{0.4712} \right) = 0.058$$

- Update Weights for Next Iteration:
 - Correct classification: Multiply by $e^{-\alpha}$
 - Incorrect classification: Multiply by e^{α}

After **Iteration 2**, the updated **weights** for the 6 samples were:

Sample	Updated Weight (Iteration 2)
1	0.1228
2	0.1374
3	0.1374
4	0.1122
5	0.1122
6	0.2311

Ensemble Techniques (AdaBoost/Adaptive Boosting)

Iteration 3: Train Third Weak Classifier

Now, using the updated weights from **Iteration 2**, we train another weak classifier.

- **New Rule:** If $X_2 < 3$, predict -1 ; else predict $+1$.

Sample	X_2	True Y	Prediction	Correct?
1	3	+1	+1	✓ Correct
2	1	+1	-1	✗ Wrong
3	2	-1	-1	✓ Correct
4	5	+1	+1	✓ Correct
5	1	-1	-1	✓ Correct
6	4	-1	+1	✗ Wrong

- **Misclassified Samples:** 2 and 6
- **Total Error (ϵ) Calculation:**

$$\epsilon = w_2 + w_6 = 0.1374 + 0.2311 = 0.3685$$

- **Compute Classifier Weight (α)**

$$\alpha = \frac{1}{2} \ln \left(\frac{1 - \epsilon}{\epsilon} \right)$$

$$\alpha = \frac{1}{2} \ln \left(\frac{1 - 0.3685}{0.3685} \right) = 0.2688$$

Ensemble Techniques (AdaBoost/Adaptive Boosting)

- Update Sample Weights
 - Correct classification: $w_{\text{new}} = w \times e^{-\alpha}$
 - Wrong classification: $w_{\text{new}} = w \times e^{\alpha}$
 - Compute new weights:

Sample	Old Weight	New Weight Formula	Updated Weight
1 (✓)	0.1228	0.1228×0.764	0.0938
2 (✗)	0.1374	0.1374×1.310	0.1799
3 (✓)	0.1374	0.1374×0.764	0.1050
4 (✓)	0.1122	0.1122×0.764	0.0857
5 (✓)	0.1122	0.1122×0.764	0.0857
6 (✗)	0.2311	0.2311×1.310	0.3022

- Normalize new weights so they sum to 1:

$$Z = 0.0938 + 0.1799 + 0.1050 + 0.0857 + 0.0857 + 0.3022 = 1.000$$

Ensemble Techniques (AdaBoost/Adaptive Boosting)

Iteration 4: Train Fourth Weak Classifier

Using the new weights, we train another classifier.

- **New Rule:** If $X_1 \geq 4$, predict $+1$; else predict -1 .

Sample	X_1	True Y	Prediction	Correct?
1	2	+1	-1	✗ Wrong
2	3	+1	-1	✗ Wrong
3	1	-1	-1	✓ Correct
4	4	+1	+1	✓ Correct
5	2	-1	-1	✓ Correct
6	5	-1	+1	✗ Wrong

- **Misclassified Samples:** 1, 2, 6
- **Total Error (ϵ) Calculation:**

$$\epsilon = w_1 + w_2 + w_6 = 0.0938 + 0.1799 + 0.3022 = 0.576$$

- **Compute Classifier Weight (α)**

$$\alpha = \frac{1}{2} \ln \left(\frac{1 - \epsilon}{\epsilon} \right)$$

$$\alpha = \frac{1}{2} \ln \left(\frac{1 - 0.576}{0.576} \right) = 0.096$$

Ensemble Techniques (AdaBoost/Adaptive Boosting)

- Update Sample Weights
 - Correct classification: $w_{\text{new}} = w \times e^{-\alpha}$
 - Wrong classification: $w_{\text{new}} = w \times e^{\alpha}$
 - Compute new weights:

Sample	Old Weight	New Weight Formula	Updated Weight
1 (✗)	0.0938	0.0938×1.101	0.1033
2 (✗)	0.1799	0.1799×1.101	0.1982
3 (✓)	0.1050	0.1050×0.908	0.0954
4 (✓)	0.0857	0.0857×0.908	0.0779
5 (✓)	0.0857	0.0857×0.908	0.0779
6 (✗)	0.3022	0.3022×1.101	0.3334

- Normalize new weights so they sum to 1.

Ensemble Techniques (AdaBoost/Adaptive Boosting)

Iteration 5: Train Fifth Weak Classifier

Using the new weights, we train another classifier.

- **New Rule:** If $X_2 \geq 4$, predict $+1$; else predict -1 .

Sample	X_2	True Y	Prediction	Correct?
1	3	+1	+1	✓ Correct
2	1	+1	-1	✗ Wrong
3	2	-1	-1	✓ Correct
4	5	+1	+1	✓ Correct
5	1	-1	-1	✓ Correct
6	4	-1	+1	✗ Wrong

Ensemble Techniques (AdaBoost/Adaptive Boosting)

Misclassified Samples: 2, 6

Total Error (ε) Calculation:

$$\varepsilon = 0.1982 + .3334 = 0.5316$$

Compute Classifier Weight (α)

$$\alpha = -0.0633$$

Update Sample Weights

- Correct classification: $w_{\text{new}} = w \times e^{-\alpha}$
- Wrong classification: $w_{\text{new}} = w \times e^{\alpha}$
- $e^{-\alpha} = e^{-(-0.0633)} = 1.0653293$

$$e^{-\alpha} = e^{-(-0.0633)} = 0.9386769$$

Sample	Old Weight	New Weight Formula	Updated Weight
1	0.1033	0.1033×1.0653293	0.110049
2	0.1982	0.1982×0.9386769	0.186046
3	0.0954	0.0954×1.0653293	0.101632
4	0.0779	0.0779×1.0653293	0.082989
5	0.0779	0.0779×1.0653293	0.082989
6	0.3334	0.1982×0.9386769	0.312955

Ensemble Techniques (AdaBoost/Adaptive Boosting)

Final Prediction Step-by-Step

Let's summarize the classifiers and their respective α values:

Iteration	Weak Classifier Rule	α
1	$X_1 \geq 3 \rightarrow +1$ else -1	0.3466
2	$X_2 \geq 2 \rightarrow +1$ else -1	0.058
3	$X_2 < 3 \rightarrow -1$, else $+1$	0.2688
4	$X_1 \geq 4 \rightarrow +1$ else -1	0.096
5	$X_2 \geq 4 \rightarrow +1$ else -1	-0.063284

Final Prediction Calculation in AdaBoost

After 5 iterations, we obtain multiple weak classifiers with different weights (α). The final prediction is determined by summing up the weighted predictions of all weak classifiers:

$$F(x) = \sum_{i=1}^5 \alpha_i h_i(x)$$

where:

- α_i is the weight of the i -th weak classifier.
- $h_i(x)$ is the prediction of the i -th weak classifier ($+1$ or -1).

Finally, the **final predicted class** is determined as:

$$\hat{Y} = \text{sign}(F(x))$$

Ensemble Techniques (AdaBoost/Adaptive Boosting)

$h_i(x)$ is the prediction of the i -th weak classifier (+1 or -1).

Apply the Sign Function

$$\hat{Y} = \text{sign}(F(x))$$

- If $F(x) > 0$, then $\hat{Y} = +1$
- If $F(x) < 0$, then $\hat{Y} = -1$

Compute Weighted Sum of Classifier Predictions

S	Y (true)	$h_1(x)$	α	$h_1(x) * \alpha$	$h_2(x)$	α	$h_2(x) * \alpha$	$h_3(x)$	α	$h_3(x) * \alpha$	$h_4(x)$	α	$h_4(x) * \alpha$	$h_5(x)$	α	$h_5(x) * \alpha$	F(x)	Final Predic tion
1	+1	-1	.3466	-.3466	+1	.058	.058	+1	.2688	.2688	-1	.096	-.096	+1	-.0633	-.0633	-0.1791	-1
2	+1	+1	.3466	.3466	-1	.058	-.058	-1	.2688	-.2688	-1	.096	-.096	-1	-.0633	.0633	-0.0129	-1
3	-1	-1	.3466	-.3466	+1	.058	.058	-1	.2688	-.2688	-1	.096	-.096	-1	-.0633	.0633	-0.5901	-1
4	+1	+1	.3466	.3466	+1	.058	.058	+1	.2688	.2688	+1	.096	.096	+1	-.0633	-.0633	0.7061	1
5	-1	-1	.3466	-.3466	-1	.058	-.058	-1	.2688	-.2688	-1	.096	-.096	-1	-.0633	.0633	-0.7061	-1
6	-1	+1	.3466	.3466	+1	.058	.058	+1	.2688	.2688	+1	.096	.096	+1	-.0633	-.0633	0.7061	1

Ensemble Techniques (AdaBoost/Adaptive Boosting)

Final Accuracy Calculation

Sample (S)	Y (true)	Final Prediction
1	+1	-1
2	+1	-1
3	-1	-1
4	+1	+1
5	-1	-1
6	-1	+1

The above table shows that samples 3, 4, and 5 were predicted correctly, while samples 1, 2, and 6 were predicted incorrectly.

$$Accuracy = \frac{\text{Number of corrected predictions}}{\text{Total Samples}} = \frac{3}{6} = 50\%$$